

# Secure and Efficient IoT Networks: An AI and ML-based Intrusion Detection System

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**Abstract**— The growing deployment of IoT devices has led to unprecedented interconnection and information sharing. However, it has also presented novel difficulties with security. Using intrusion detection systems (IDS) that are based on artificial intelligence (AI) and machine learning (ML), this research study proposes a unique strategy for addressing security issues in Internet of Things (IoT) networks. This technique seeks to address the challenges that are associated with these IoT networks. The use of intrusion detection systems (IDS) makes this technique feasible. The purpose of this research is to simultaneously improve the present level of security in ecosystems that are connected to the Internet of Things (IoT) while simultaneously ensuring the effectiveness of identifying and mitigating possible threats. The frequency of cyber assaults is directly proportional to the increasing number of people who rely on and utilize the internet. Data sent via a network is vulnerable to interception by both internal and external parties. Either a human or an automated system may launch this attack. The intensity and effectiveness of these assaults are continuously rising. The difficulty of avoiding or foiling these types of hackers and attackers has increased. There will occasionally be individuals or businesses offering IDS solutions who have extensive domain expertise. These solutions will be adaptive, unique, and trustworthy. IDS and cryptography are the subjects of this research. There are a number of scholarly articles on IDS. An investigation of some machine learning and deep learning techniques was carried out in this research. To further strengthen security standards, some cryptographic techniques are used. Problems with accuracy and performance were not considered in prior research. Furthermore, further protection is necessary. This means that deep learning can be even more effective and accurate in the future.

**Keywords**— Security, IoT Networks, AI, ML, IDS

## I. INTRODUCTION

In response to the growing need for cyber security, a number of research investigations have been carried out in order to classify the information that is sent over an IoT network [1]. AI and ML-powered intrusion detection is a method used to identify and mitigate network hazards by detecting any unauthorized individuals attempting to gain access. Deep learning techniques have been used in the present research work, which has explicitly investigated the need for intrusion detection and the strategy for its implementation [2].

### A. Cybersecurity

Cybersecurity refers to the process of safeguarding computer systems, computer networks, and computer programs against digital attacks. These breaches often aim to achieve one of three objectives such as obtaining sensitive

information, altering, erasing, or disrupting regular company operations [3]. Implementing effective cybersecurity measures in today's environment is particularly challenging due to the proliferation of connected devices, which already surpasses the human population, and the increasingly innovative tactics used by cybercriminals. In order to bolster cyber security, we may use the IDS [4], which is a very renowned security framework designed to counteract a wide range of intrusion threats.

### B. Intrusion Detection System

In other words, it is a monitoring system that is meant to detect and identify any unexpected behavior. When it recognizes actions that are deemed to be suspicious, it also sends out alerts [5]. Given these signals, security experts have the ability to thoroughly examine the problem. Subsequently, security analysts proceed to implement the necessary measures to resolve the danger [6]. Essentially, it transforms into an intrusion-capturing model. This model is often referred to as the Intrusion Detection Model, and it has been created [7] as shown in Figure 1.

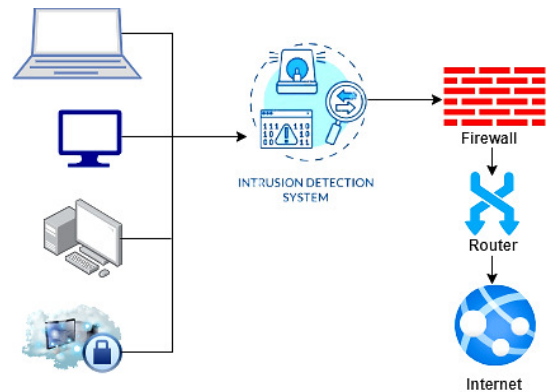


Fig. 1. Intrusion Detection System

### C. Cryptography

Cryptography is a discipline that concentrates on the advancement and examination of diverse cryptographic [8] procedures and approaches to guarantee the confidentiality of message contents between the sender and the receiver [9]. The term "cryptic" originates from the Greek word "Kryptos". Encryption is the act of transforming plain, readable material into a coded form [10], while decryption is the act of reversing this process to get the original content. Cryptography refers to the process of encoding and decoding data [11]. Cryptographies are vital for safeguarding sensitive information from unauthorized access [12, 13, 14].

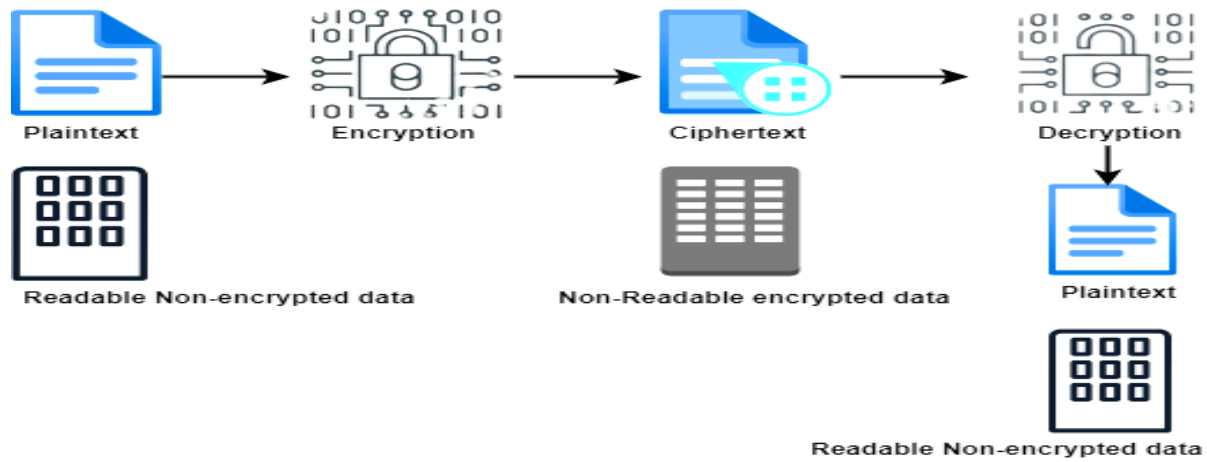


Fig. 2. Cryptography

Cryptography cannot prevent hackers from compromising or accessing data [15]. Cryptography ensures only the intended receiver can decipher encrypted data. External sources did not provide the content to unauthorized parties [16]. The term "crypt" means secrecy, whereas "graphy" means writing. Data can be encrypted while being stored and transported [17] to prevent malicious parties from accessing or altering it [18]. By confirming sender and receiver legitimacy, this strategy may decrease retracted messages. Figure 2 shows how cryptographical process works.

#### D. Role of Cryptography In IDS using Security Enhancement

Cryptography encrypts data so only a trusted decryptor can read it [19]. Cryptography is essential in modern computer systems because it securely and cheaply protects data. IDS software monitors and alerts appropriate parties to potentially dangerous actions [20]. These signals can help SOC analysts and incident responders assess a danger and choose the optimal response [21]. Regular polling and encryption protect the communication route and data. When an alarm is enabled, only the specified receiver may decrypt the encrypted message [22]. If traffic is tunneled into an encrypted connection, the IDS cannot assess or notify. IDS may notify for malicious network traffic [23]. It is a software program that monitors a network or computer system for malicious activity [24], such as rule violations. The IT personnel can fix these issues [25].

## II. LITERATURE REVIEW

The advancement of intrusion detection systems (IDS) has been significantly aided by a number of studies that have been conducted to investigate deep learning (DL). This research study used several datasets to provide varying degrees of precision. Within the scope of a research project that was carried out by Y. H. Choi (2020), the datasets UNSW-NB15 and CICIDS2017 were used for the objectives of training and classification [1]. Following the implementation of deep learning processes, the scientific investigation was finished. In their study, S. Al-Emadi (2020) used deep learning approaches to identify intrusions using the NSL-KDD dataset [2]. In their study, Hao (2020) used the Artificial Neural Network-Based Sequential Detection Method in order to create an intrusion detection system that is tailored to the KDD99 and NSL-KDD datasets. The research yielded a level of accuracy below 90% [3]. In their study, Ashiku (2021) proposed IDS for networks that used DL methodology [4].

As part of the research carried out in 2021, A. Thakkar conducted a thorough analysis of machine learning and deep learning models. This inquiry was in conjunction with the development of intrusion detection systems on the Internet of Things (IoT). They made use of the datasets that DARPA and KDD CUP 99 provided while they were in deep learning. As a result, they were able to attain an exceptional accuracy rate of more than 90 percent [5]. In 2021, S. Varshney offered a system that was designed for the purpose of deploying an intelligent intrusion detection system. Using KDD and NSL datasets to train computer learning models, this system is working toward its objective of achieving an accuracy rate that is more than 95% [6]. In order to accomplish this aim, the system needs training data. In 2021, S. W. Lee conducted a thorough investigation and study to ensure the security of IDS [7]. In a comprehensive study, Z. Ahmad (2021) conducted a thorough examination of machine learning in conjunction with deep learning methodologies. The accuracy achieved by this deep learning method was less than 98% [8]. In a study conducted by M. R. Ahmed in 2022, a thorough investigation was carried out on intrusion detection systems with an accuracy of 95% [13]. [26] suggested solution based on Pyspark-ML. Table 1 presents their purpose, technique, results, and limitations.

## III. PROBLEM STATEMENT

Various research studies and questions have been conducted on the subject of intrusion detection systems (IDS). The conventional study included several machine learning (ML) and deep learning (DL) approaches for conducting intrusion detection. Moreover, there are several cryptographic techniques available to increase security. Previous studies have neglected the difficulties related to performance and accuracy. Furthermore, it is necessary to improve the security mechanism. Therefore, more research is needed to improve the precision and effectiveness of deep learning applications. Numerous research projects are now dedicated to intrusion detection systems. However, there is a recognized need to improve the efficiency and accuracy of conventional research investigations concerning IDS. It has been observed that there is a need for more work to improve scalability and flexibility. The proposed approach employs encryption and intrusion detection systems (IDS) to improve security.

TABLE I. REVIEW OF LITERATURE

| Ref  | Author       | Year | Objective                                                                                                                                                                         | Dataset                                                               | Methodology                                                                                  | Result                              | Limitation                                                                         |
|------|--------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------|------------------------------------------------------------------------------------|
| [3]  | Z. Hao       | 2020 | The Application of Artificial Neural Network Sequential Detection in Intrusion Detection Systems                                                                                  | KDD99 and NSL-KDD                                                     | ANN                                                                                          | Accuracy<90%                        | Research is being done on both ANN and IDS.                                        |
| [4]  | L. Ashiku    | 2021 | Network Intrusion Detection System that is Based on Deep Learning                                                                                                                 | UNSW-NB15                                                             | Deep Learning                                                                                | accuracy is about 95-96%            | There is a significant amount of focus placed on both DL and IDS.                  |
| [5]  | A. Thakkar   | 2021 | Examining the IDS from both a machine learning and deep learning perspective for the Internet of Things                                                                           | DARPA and KDD CUP 99                                                  | ML, IoT, DL, IDS                                                                             | Accuracy>90%                        | Certain restrictions are imposed on the level of performance that may be achieved. |
| [8]  | Z. Ahmad     | 2021 | A comprehensive investigation of the use of machine learning and deep learning for the purpose of developing network intrusion detection systems                                  | Both the CSE-CIC-IDS2018 benchmark dataset and the KDD Cup'99 dataset | Deep learning                                                                                | Accuracy<98%                        | A greater amount of work is required of them in order to guarantee accuracy.       |
| [13] | M. R. Ahmed  | 2022 | Through the use of machine learning and deep learning, a comprehensive analysis of intrusion detection systems in software-defined networks is presented.                         | A Benchmark Dataset, Particular to the SDN                            | Deep learning, machine learning, and software-defined networks are all examples of studies.  | Accuracy lies between 90 and 95%    | According to this study, the technological aspect is not being worked on.          |
| [17] | J. Jose      | 2023 | To build intrusion detection systems for the internet of things, deep learning will be used to the CIC-IDS 2017 dataset. This will allow for identification of potential threats. | CIC-IDS 2017 dataset                                                  | The Internet of Things Network Intrusion Dataset is made available using the IEEE Data Port. | DNN=94.61%, LSTM=97.67%, CNN=98.61% | Limited accuracy                                                                   |
| [23] | Bani-Salameh | 2021 | A model for the prediction of priorities that is based on deep learning.                                                                                                          | JIRA dataset with 2000 bug reports                                    | SVM, LSTM                                                                                    | LSTM: 0.898<br>SVM : 0.865          | Limited performance                                                                |

#### IV. PROPOSED WORK

Neural network models play a significant role in intrusion detection systems (IDS). Supervised learning involves neural networks learning from datasets used in intrusion detection systems. They refine and enhance the accuracy of their predictions to identify the correct answer. In addition to their widespread use in fields like market research and sales forecasting, neural networks find use in network security. While Figure 3 shows the conventional method, the proposed

method uses an LSTM model after dataset preprocessing. During the process of orchestrating the categorization of the model's data, a multitude of components are given careful

attention. These elements include the complex interaction between the activation function, dropout mechanism, fully connected layer, and the subtle dynamics of the long-short-term memory layer. Each of these components makes a distinct contribution to the model's capacity to recognize patterns, adjust to varied degrees of complexity, and eventually reveal the underlying insights that are concealed inside the data environment.

The IDS under discussion uses LSTM networks. LSTM and fully linked layers must be used during training. The dataset is partitioned to train the LSTM model using the attributes. Batch size affects training time and outcome accuracy. There are numerous more features, like dropout and

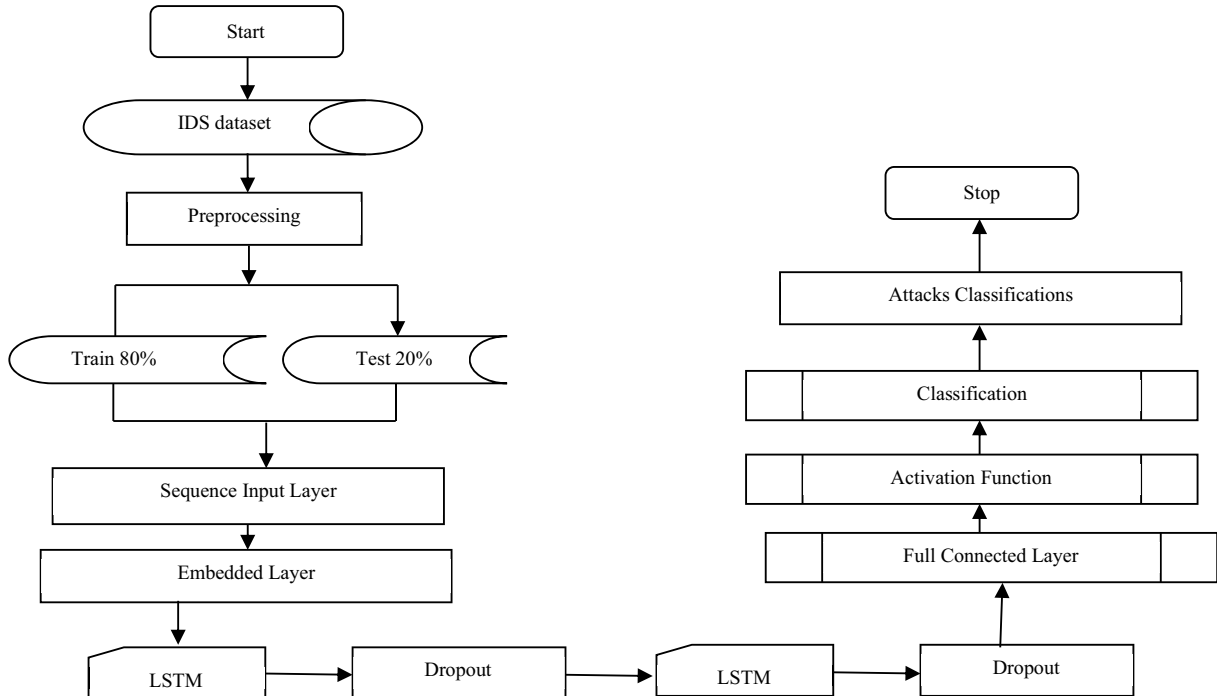


Fig. 3. Proposed model by Integration of LSTM with IDS Model

masked layers. Dropout layers are excellent for overfitting. LSTM procedures analyze IDS data and make predictions in the LSTM layer. The model has two layers with opposing relationships built using LSTM. The degree of concealment varies per layer. Over longer training cycles, LSTM layers improve gradient flow and develop long-term dependencies.

- The term "overfitting" refers to the occurrence that takes place when the model gets too customized to the training parameters. In order to eliminate this problem, the dropout layer is used.
- Creating the output of a fully connected layer involves first conducting matrix multiplication between the input and the weight matrix of the layer, and then adding a bias vector to the result. This process is repeated until the output is formed.
- This procedure is carried out several times till the results are finished. It is necessary to continue with this operation until it is finished in order to ensure that the result is created accurately.
- One of the functions of softmax is to take responsibility for triggering the final layer. This function is responsible for transforming the output of the layer into a probability distribution.
- The testing step entails assessing the trained model on a portion of the whole dataset.
- When it comes to quality assurance, this is the stage when the trained model is put through its paces. The accuracy of the model may be evaluated with the assistance of the example dataset, which is available.

One of the networks that is taken into consideration during the evaluation phase is one that has been trained under supervision. Following this, we are provided with the test dataset. An evaluation of a trained network on the test data, taking into consideration fresh test values, is used to calculate the metrics of accuracy, F-score, and precision throughout the computation process.

## V. RESULT AND DISCUSSION

Our sophisticated intrusion detection system (IDS) is built on UNSW-NB 15, a massive collection of unfiltered network packets carefully evaluated using IXIA PerfectStorm software. Our strategy is based on this resource. This project aimed to create an environment where artificially generated and naturally occurring antagonistic behaviors might coexist. Researchers utilized Tcpcap to collect 100 terabytes of unprocessed Pcap data. This article covers nine types of assaults, including fuzzy logic, analysis, denial of service, shellcode, and worms. As we studied experimental IDS approaches, the UNSW-NB 15 dataset became our main focus. UNSW-NB 15 is the newest dataset in our library, compared to the NSL KDD 99, the oldest. This highlights it as the latest contribution. CAIDA and ISCX are newer datasets. This extensive investigation covers recent limited-effect assaults and other network activity, including weird and normal behaviors. The dataset is useful for evaluating national intrusion detection systems (NIDS) since it is well-organized and free of duplicate data. The dataset comprises 2,540,044 entries in four.csv files. These entries are divided into three categories. Training entries total 175,341, while testing entries total 82,332. Figure 4 shows that most of the training set examples are attacks, whereas just one-third are conventional

transactions. Figure 5 shows a striking distribution of data from this examination: Attack traffic makes up 55.1% of the test set, while routine transactions make up 44.9%. The dataset's class labels employ numerical encoding to designate assaults and normal traffic as 1 and 0. The data may be condensed into these numbers.

Transaction class distribution of UNSW NB-15 Train Dataset

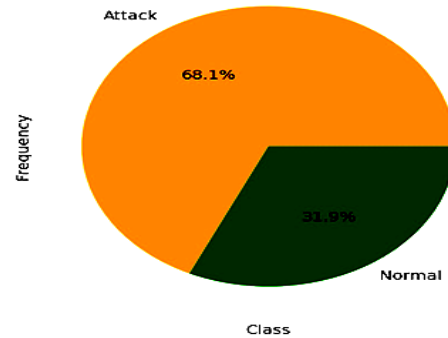


Fig.4. Statistical breakdown of the transaction types included within the UNSW-NB15 training dataset.

We input our network model from training with one dataset into other datasets to produce predictions. The testing step proves the model's reliability, allowing assessment. A supervised-trained network is considered for assessment. Following that, we get the test dataset. Accuracy, F-score, and precision are calculated by evaluating a trained network on test data with new test values. We'll use 80% of the data to train the model and 20% for testing..

Transaction class distribution of UNSW NB-15 Test Dataset

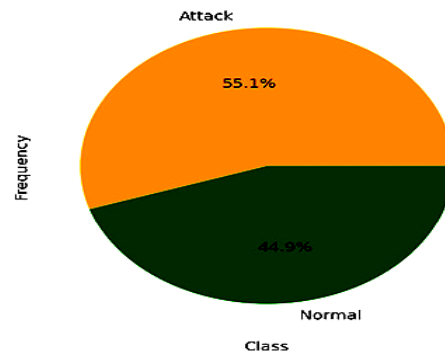


Fig. 5. Within the test dataset of UNSW-NB15, the distribution of transaction types is shown.

The model was 96.76% accurate for IDS. Use the confusion matrix to evaluate the model classification system's data handling. The confusion matrix can help. The dataset subset has 5,250 samples. In those samples, 4,974 were negative and 4,174 were positive. This indicates a high degree of identification accuracy. Intrusions were wrongly categorized as non-intrusions 233 times, while non-intrusions were incorrectly classed as intrusions numerous times. Table 2 shows the categorization report's model efficacy assessment. The evaluation is part of it. The resemblance between actual events and expected outcomes can be used to evaluate a forecast's accuracy. The forecast may be used to produce more accurate forecasts. This ensures that the prediction can be used to get accurate forecasts. The recall statistic can count positive predictions that came true. This has major benefits. Table 2 shows that the model compares harmful and benign network

data with appropriate accuracy and performance. Classification report Model accuracy & loss during training and validation has been shown in Figure 6 and Figure 7 respectively.

TABLE II. OVERALL ACCURACY AND PERFORMANCE

|              | Precision | Recall | F1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.96      | 0.99   | 0.97     | 5047    |
| 1            | 0.98      | 0.95   | 0.96     | 4407    |
| Accuracy     | 0.96      | 0.96   | 0.97     | 9454    |
| Macro avg    | 0.97      | 0.97   | 0.97     | 9454    |
| Weighted avg | 0.97      | 0.97   | 0.97     | 9454    |

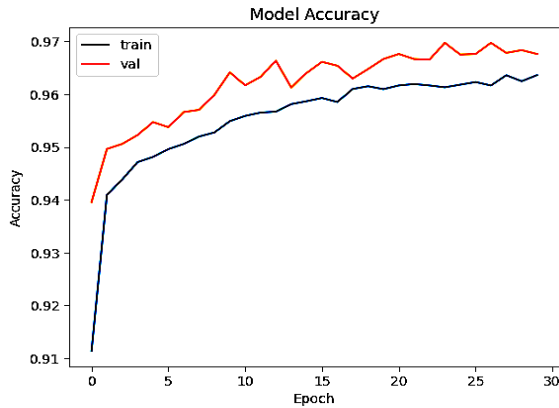


Fig. 6. Model of Accuracy

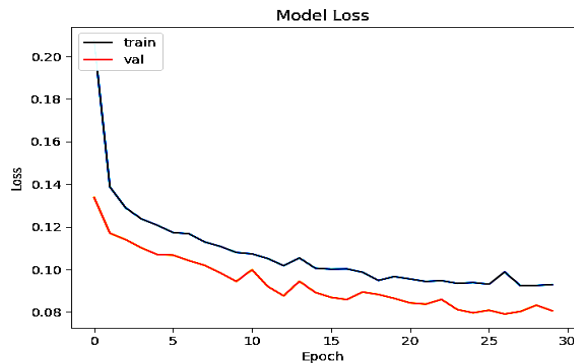


Fig. 7. Model of Loss

A comparison of the work that has been suggested to the work that has been done in traditional research has been made. The work that has been suggested has been seen to provide a higher level of accuracy in comparison to the work that has been done traditionally.

TABLE III. ACCURACY COMPARISON

|          | ANN[3] | Conventional deep learning approach[4] | DNN[17] | LSTM [17] | Proposed work |
|----------|--------|----------------------------------------|---------|-----------|---------------|
| Accuracy | 90%    | 95%                                    | 94.61%  | 97.67%    | 98%           |

Considering results of Table 3, Figure 8 has been generated to show comparative analysis.

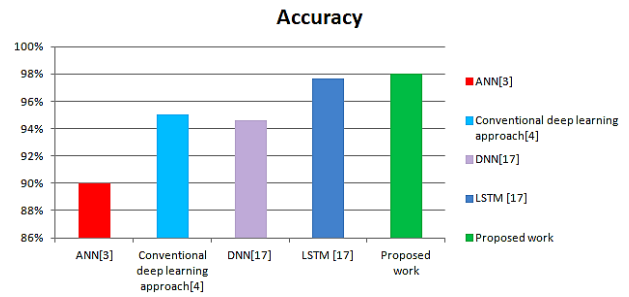


Fig. 8. Accuracy Comparison

F1-score is aggregate of recall and precision and following table presents comparison of F1-score in case of proposed and conventional work.

TABLE IV. F1-SCORE COMPARISON

|          | Conventional deep learning approach[4] | LSTM [17] | Proposed work |
|----------|----------------------------------------|-----------|---------------|
| F1-Score | 95%                                    | 96%       | 98%           |

Considering Table 4, F1 –score comparison has been shown in following Figure 9.

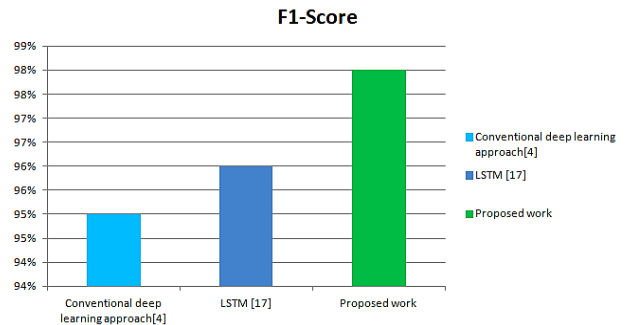


Fig. 9. F1-Score Comparison

## VI. CONCLUSION

The proposed IDS utilizes AI and ML techniques to evaluate the extensive volumes of data produced by IoT devices in real-time. The system may detect aberrant actions that may indicate prospective intrusions by using methods such as anomaly detection and pattern recognition. By using machine learning models, the IDS is able to dynamically adjust and develop, consistently enhancing its capability to differentiate between regular and malicious behaviors. The study focuses on doing a thorough examination of the current security concerns in IoT networks, with a particular emphasis on the distinct vulnerabilities that arise from the wide variety of IoT devices. The design of IDS is explained, emphasizing the use of advanced algorithms to promptly and precisely identify potential threats. Additionally, the paper addresses the efficacy of the strategy that was recommended, stressing the fact that it has a low incidence of false-positive findings and has a minor effect on the performance of the network. Case studies and simulation results highlight the effectiveness of the AI and ML-based intrusion detection system (IDS) in real-world Internet of Things (IoT) scenarios.



## VII. FUTURE SCOPE

Subsequent investigations will greatly influence the approach used to provide dependable IDS predictions. The study has developed an adaptable and expandable strategy that incorporates the training model. The proposed model is expected to derive advantages from its comprehensive training on a large dataset, resulting in improved accuracy. Continuing this line of investigation is essential for improving IDS detection. The study's findings will have significant consequences for enhancing our capacity to forecast IDS. It is possible that comparable approaches may be used in future studies to provide a more accurate outcome for IDS detection. By using the optimization approach in future investigations, it may be possible to act as a catalyst for sharpening datasets, which will ultimately result in an increase in the effectiveness of the intrusion detection system (IDS). In addition, the exploration of more complex classifiers and the exploration of deep learning models might open up new doors for the improvement of operational efficiency and the attainment of levels of accuracy that have never been seen before.

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